

High litter moisture content suppresses litter ammonia volatilization¹

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ABSTRACT With global food demand expected to increase by 100% in the next 50 yr, urgency to combine comprehensive strategies for sustainable, efficacious, and environmentally sensible agronomic practices has never been greater. One effort for US meat bird management is to reduce NH₃ volatilization from litter to create a better growing environment for the birds, improve production efficiency, retain N in litter for fertilizer value, and negate the detrimental environmental impacts of NH₃ loss to the air. To derive the fundamental effects of temperature and moisture on litter NH₃ volatilization over the range of conditions found in commercial houses, experiments were conducted using commercial broiler litter that had moisture contents of approximately 20 to 55% while controlling temperatures ranging from 18.3 to 40.6°C. Litter samples (100 g) were placed in 1-L containers that received humidified air at approximately 113 mL/min. Volatilized NH₃ in exhaust air was captured in H₃BO₃ traps. Ammonia

loss (log₁₀ transformation) was modeled via an equation using linear coefficients for temperature and moisture, an interaction term for temperature × moisture, and a quadratic term for moisture. The surface responses resembled parabolic cylinders, indicating a critical moisture level at which NH₃ no longer increases but is diminished as moisture continues to increase. The critical moisture level lies between 37.4 and 51.1% litter moisture, depending on the temperature. An increase in temperature consistently increased NH₃ generation. When the temperature extremes were compared, the maximum NH₃ was up to 7 times greater at 40.6 vs. 18.3°C. The upper moisture limit at which NH₃ release is maximized and subsequently arrested as moisture continues to increase had not been defined previously for commercial broiler litter. The poultry industry and researchers can use these results as a decision tool to enable management strategies that limit NH₃ production.

Key words: ammonia, broiler, litter, moisture content, temperature

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INTRODUCTION

Until the last decade, much of the published broiler NH₃ emission data were from European studies that lacked relevance to US climatic conditions and management practices. The few models generated in recent research in the United States do not have proven applicability across commercial facilities. To exacerbate the unpredictability of quantifying NH₃ emissions, the contract grower structure of broiler production in the United States allows management practices that are subject to individualistic preferences. The emphasis on studies quantifying emissions is primarily a response to impending regulations on animal production facilities.

Annually, animal feeding operations release large quantities of NH₃, accounting for 50 to 70% of the total release in the United States (NRC, 2003).

Fundamental effects on NH₃ release from broiler litter have been reported previously, in order of ascending influence, as moisture, temperature, and pH (Elliott and Collins, 1982; Carr et al., 1990). Elliott and Collins (1982) reported that either extremely low or high litter moisture contents reduced NH₃ volatilization, but they found no valid correlation between litter moisture content and NH₃ levels for intermediate moisture contents. Because litter pH is normally greater than 8, Coufal et al. (2006) cited temperature and moisture as the 2 most important variables affecting NH₃ volatilization in commercial facilities. In commercial broiler houses, temperature and moisture may be affected by a host of factors: housing design and management, bird age and health status, drinker management and maintenance, litter age and management, and ambient weather conditions (Coufal et al., 2006). In a single winter flock, litter surface temperatures have been reported to range

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from 18 to 42°C, with corresponding litter moisture levels of 12 to 60% (Miles et al., 2008b). These ranges of temperature and moisture include extremes because of external conditions (e.g., low litter temperatures in the nonused portion of the facility during half-house brooding), management practices (brood heaters causing hot, dry litter), and bird effects (high litter moisture in caked areas). Capturing extremes lends greater applicability to a variety of house environments in the United States.

The objective of this work was to model NH₃ discharge from built-up broiler litter at gradient litter moisture levels and over the range of litter temperatures within commercial houses. The expected outcome was to provide predictions of changes in NH₃ volatilization with visualization via surface response plots for temperatures between 18.3 and 40.6°C (65 to 105°F) and litter moisture levels from 20 to 55%.

MATERIALS AND METHODS

Experimental Design

To determine the relative NH₃ discharge for commercial broiler litter as a function of temperature and moisture, a factorial arrangement of treatments was employed, using 5 temperatures (18.3, 23.9, 29.4, 35, and 40.6°C) and 5 moisture contents (initial, 25, 35, 45, and 55%) measured over 4 d. Because only 3 climate chambers were available for temperature treatments, a balanced incomplete block (BIB) design was used, with the block as the time for 3 of the 5 temperatures. The temperature treatments were assigned in the BIB design (plan 11.1a, page 471, Cochran and Cox, 1967), which had 10 blocks. Each temperature treatment was tested 6 times in the 10 blocks, and each possible pairing of 2 temperatures occurred 3 times (Table 1). The efficiency of this design is not less than 83% of the randomized complete block design with the same number of observations (Cochran and Cox, 1967). Within each experiment and each incubator, the 5 moisture treatments were arranged in a randomized complete block design with 2 subsamples.

NH₃ Measurement System

The method for determining NH₃ generation from litter was described previously (Miles et al., 2008a). For

the current gradient moisture–temperature study, a second chamber acid trap (CAT) system was developed by constructing 3 additional manifolds and mounting each one within an incubator or environmental chamber (pH Environmental, Division of Harris Manufacturing Co., North Billerica, MA). The manifolds supply water-scrubbed air at a flow rate of approximately 110 mL/min to twelve 1-L chambers (containers) in each incubator. After moisture treatments were prepared with commercial litter (described below), 100-g litter samples were weighed into each 1-L chamber. Samples were distributed across the bottom of the chambers so that the litter occupied the entire chamber area (93.8 cm²) and to provide a uniform surface area for NH₃ volatilization. The flow rate was achieved by maintaining the main air pressure at 82.7 kPa (12 psig) and each manifold at approximately 34.5 kPa (5 psig). The overall average air flow rate for the system was 112.6 mL/min, with an SD of 5.5 mL/min.

Exhaust air from each container flows through a series of 2 H₃BO₃ traps (flasks containing 30 mL each), where NH₃ is captured. The combined trap contents were titrated with HCl at 24-h intervals for 4 d to determine milligrams of N volatilized from the litter. The time periods for titration (each 24 h) and the overall length of an experiment (4 d) were chosen based on previous experience.

Litter Preparation

Commercial Litter Sampling. Broiler litter was obtained from a commercial farm in Mississippi, where the original bedding material was pine wood chips. The farm operates 8 tunnel-ventilated, solid-sidewall houses, 4 of which were built in 2003 and 4 of which were built in 2005. The flocks that began in August 2008 were grown on litter that had previously grown 36 flocks in the older houses and 17 flocks in the newer houses. Litter for each test came from a single house. The research plans included the use of litter from a newer house. An unexpected total cleanout of that house required that litter was then sampled for 2 of the blocks from one of the older houses. In the first sampling of the older house, the litter had been meagerly topped with rice hulls, providing the only (assumed negligible) exception to the original bedding being pine wood chips. Litter sampling was conducted approximately 3 d before

Table 1. Balanced incomplete block experimental layout¹

Temperature (°C)	Block									
	1	2	3	4	5	6	7	8	9	10
18.3	×	×	×			×	×	×		
23.9	×	×		×		×			×	×
29.4	×			×	×		×	×	×	
35.0			×	×	×	×	×			×
40.6		×	×		×			×	×	×

¹Cochran and Cox (1967).

Table 2. Collection data and initial characteristics of commercial litter

Block	Flock; litter characteristic	Date collected	Moisture (%)	pH	Total N (%)
1	17; 3 d after sold at 54 d, before decake	August 2008	25.6	8.43	3.95
2	17; after decake	August 2008	20.5	8.28	3.66
3	18; d 8	September 2008	16.9	8.13	3.93
4	18; d 16	September 2008	18.9	8.11	4.18
5	18; d 23	September 2008	20.6	7.73	4.03
6	40; after rowing and topped with rice hulls, sold at 46 d	May 2009	17.8	7.93	3.45
7	41; d 36	June 2009	31.2	8.70	3.63
8	2; end of flock, sold at 42 d	September 2009	30.1	8.47	3.06
9	3; d 6	September 2009	11.5	8.23	2.00
10	3, d 27	October 2009	21.9	7.80	3.24

each experiment began to accommodate pretest moisture analysis and to apply moisture treatments. Weekly sampling for each block maintained the litter closest to its natural, albeit dynamic, environment before applying moisture treatments. Table 2 lists the initial litter characteristics (percentage of moisture, pH, and percentage of N) for the 10 experiments as well as the litter age (within and over all flocks) and collection dates. The bulk litter sample was collected at a depth of no more than 7.6 cm (3 in.) from a 61 × 91 cm (24 × 36 in.) area in the center-nonbrood end of the house. In the laboratory, the gross sample was thoroughly mixed before subsampling for physical and chemical analyses. The bulk sample was loosely covered and refrigerated until laboratory sample preparation.

Laboratory Sample Preparation. The day before the experiment start date, litter was weighed into 10 separate 450-g portions for creating the moisture treatments (25, 35, 45, 55%, 2 of each) and to maintain 2 sources of the initial litter sample. When the initial (original) moisture for the sample was near 25% (e.g., within approximately 3%), no 25% level was created because accurate mixing with distribution of a small amount of H₂O was not possible. Deionized H₂O was added to the 450-g portions based on the moisture gradient spreadsheet calculation (not shown). For each moisture content of the original litter sample, the spreadsheet calculated the H₂O added to create a sample with the target moisture. Water was weighed in a beaker and then poured over the sample and mixed with a gloved hand until homogeneous. The litter-moisture mixtures as well as the original litter samples were placed in gallon Ziploc bags, the excess air was removed, and the mixtures were refrigerated until the experiment was started (approximately 24 h later).

Physical and Chemical Analyses. After the samples were returned to the laboratory, subsamples of the homogenized bulk litter were oven-dried (65°C for 48 h) to determine moisture so that the gradient moisture treatments could be calculated. Before and after each trial, litter moisture was found by the loss in weight (same conditions as above), and pH was measured by using a deionized H₂O-to-litter sample ratio of 5:1. Moisture and pH were determined for all treatment combinations (gradient moisture and temperature).

To avoid measuring a dilution effect and for cost effectiveness, only the original litter samples from each experiment were analyzed for total N, NH₄⁺, and NO₃⁻. These included 1 sample from the beginning and 3 samples at the end (1 at each of the 3 temperatures) of each experiment. All analyses were performed in duplicate. Total N in each treatment (as-is basis) was analyzed by direct combustion (CE Elantech, Lakewood, NJ) using 40-mg subsamples. In addition, the original and end litter samples were extracted for NH₄⁺ and NO₃⁻ by using 2 g of sample extracted with 20 mL of 2 M KCl and shaken for 1 h (Keeney and Nelson, 1982). After the extraction, the samples were filtered through a funnel containing Whatman number 1 filter paper and put into labeled scintillation vials, then acidified with 1 drop of 2 M HCl and frozen until analysis for NH₄⁺ and NO₃⁻. Flow injection analysis (QuikChem 8000, Lachat Instruments, Milwaukee, WI) was used to determine NH₄⁺ and NO₃⁻ of the original and ending litter samples. The results of these analyses are given in Table 3.

Initial and final total N were used with the measurement of captured NH₃ to determine the N mass balance for each incubator in each of the 10 experiments. The N mass balance for the study assumed the initial total N in the litter was equal to the sum of the final total N in the litter plus the cumulative NH₃ lost. The values for each (Table 3) were converted to milligrams of N per kilogram of litter to calculate the percentage accounted for as (final/initial) × 100.

Statistical Analyses

Procedures of SAS (SAS Institute, Inc., 2003) were used in the following analyses. The first step in evaluating responses for NH₃ volatilization (mg of N) for the gradient moisture-temperature experiments was to check normality using the PROC UNIVARIATE procedure. The distribution of the response variable was found to be positively skewed. A log₁₀ transformation reduced the skewness to a very acceptable value near zero (-0.25). This result indicated the need to use log₁₀ of NH₃ transformation on all further analyses.

A repeated measures ANOVA was executed using the PROC MIXED procedure. The covariance structure

Table 3. Results of commercial litter N analysis and mass balance

Block	Incubator temperature (°C)	NH ₄ ⁺ (mg/kg)		NO ₃ ⁻ (mg/kg)		Total N (%)		Cumulative NH ₃ loss (mg of N/kg)	Percentage accounted for
		Start	End	Start	End	Start	End		
1	18.3	6,933	7,200	256	222	3.95	3.88	519	99.6
	23.9		6,703		179		3.69	731	95.4
	29.4		6,793		168		3.65	1,017	94.9
2	18.3	5,990	6,217	165	149	3.66	3.73	245	102.4
	23.9		6,282		156		3.79	326	104.3
	40.6		7,110		134		3.73	777	103.9
3	18.3	8,092	9,033	172	185	3.93	3.60	79	91.8
	35		8,630		177		3.85	333	98.7
	40.6		8,432		160		4.13	440	106.2
4	23.9	9,015	8,051	198	174	4.18	3.95	139	94.8
	29.4		8,304		181		3.91	219	94.0
	35		8,971		177		3.92	294	94.6
5	29.4	8,129	8,304	190	170	4.02	4.12	328	103.2
	35		8,919		169		4.41	415	110.6
	40.6		8,338		167		4.21	467	105.9
6	18.3	4,609	4,343	10,752	10,755	3.45	3.32	230	96.8
	23.9		4,047		10,114		3.50	316	102.4
	35		3,976		10,373		3.66	750	108.2
7	18.3	5,848	5,780	1,133	1,180	3.63	3.15	681	88.5
	29.4		3,856		998		3.31	1,668	95.7
	35		3,442		1,069		3.31	1,916	96.4
8	18.3	7,114	2,173	745	562	3.06	3.17	837	106.3
	29.4		5,547		697		3.00	1,677	103.4
	40.6		3,824		701		3.01	2,420	106.1
9	23.9	1,925	2,090	3,128	3,261	2.00	2.15	85	107.7
	29.4		1,987		3,558		2.28	120	114.6
	40.6		1,869		3,297		2.26	257	114.2
10	23.9	3,635	2,633	5,172	4,558	3.24	2.66	336	83.2
	35		2,142		4,084		2.98	611	93.9
	40.6		2,114		3,627		2.32	708	73.8

was modeled using 3 candidate structures: a compound symmetry model, an autoregressive integrated moving average covariance model, and a Toeplitz 2-level model. The best covariance structure was determined using the Akaike information criteria (Akaike, 1974). The model that minimized the Akaike information criteria (Littell et al., 2006) was the compound symmetry covariance. An implication of compound symmetry covariance is that random variables are equally correlated and have equal variances (Milliken and Johnson, 1998). When the repeated measures ANOVA indicated that an interaction was significant, the interaction was investigated by analyzing responses for 1 or 2 other factors at each level of another factor. Because the CAT system causes sample drying over time without additional moisture inputs, it is appropriate to consider daily responses separately.

Covariance analysis (PROC MIXED) was used to determine simple and complex regression responses for the quantitative variables (induced temperature and moisture conditions). Where appropriate, the significance of linear and quadratic responses and their interactions were determined for all quantitative variables. Under the principle of parsimony, a minimal number of significant variables was used in regression equations describing NH₃ generation. When 2 quantitative variables interacted, they were analyzed as response surfaces, and figures were generated to assist in the interpretation of those complex responses. A MATLAB (2008) program

was written to use the resulting regression equations and coefficients to develop the surface response plots.

RESULTS AND DISCUSSION

Litter Characterization

For a thorough understanding of the NH₃ volatilization model described later, supporting information is first given to characterize how the experiment induced changes in the initial litter samples. Figure 1 shows the litter moisture as a deviation from the target percentage. The moisture content was within 0.03 percentage points of the desired level, with the direction of deviation almost balanced. The nonhomogeneous nature of the litter components (feces, feed, and pine chips) could have contributed to this moisture variability, along with the variable composition of particle sizes.

The 3-d storage before start of the test is another factor that may have affected moisture level. Between collecting the litter and creating the moisture treatments, the litter was stored refrigerated and loosely covered. Refrigeration was chosen to slow NH₃ release from the litter. The bulk sample was not sealed to ensure no buildup of NH₃ within the container, which could have had an adverse effect on the microbial constituents in the litter. Figure 2 shows the change in initial moisture between the collection and start of the test. Comparing the percentage of moisture in the original sample with

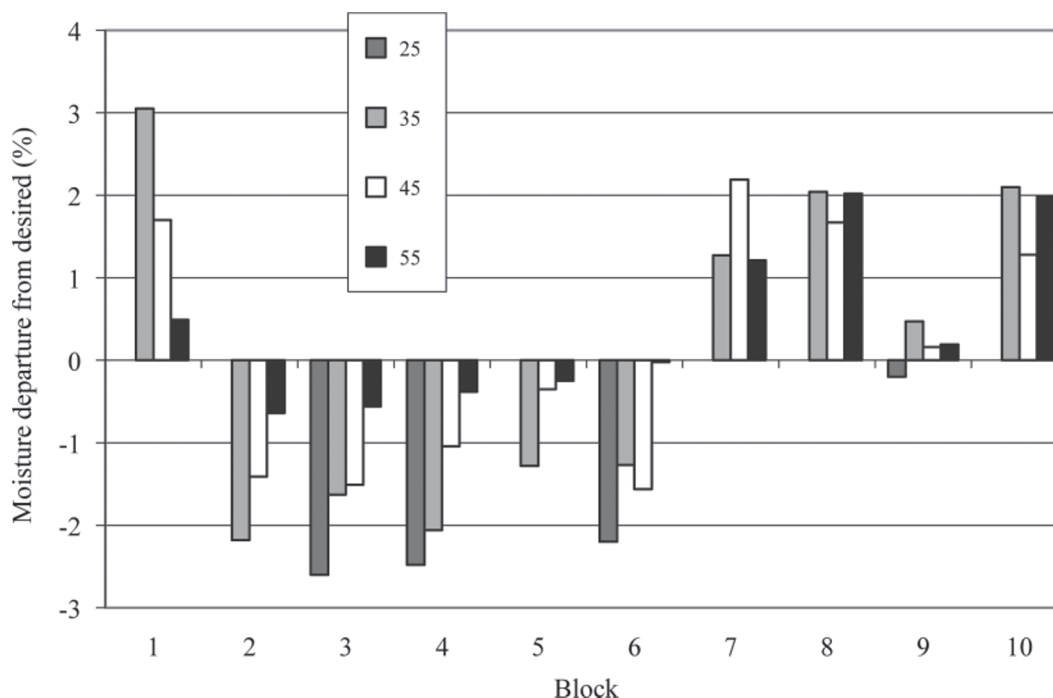


Figure 1. Moisture departure from the desired level (25, 35, 45, or 55%) for commercial broiler litter in each block.

the actual percentage at the start of the test indicates that in blocks 1 to 6 and 9, the samples lost moisture during the time period before the test actually began. The loss of moisture before starting the test (Figure 2) did not consistently correspond to litter treatments drier than desired (Figure 1). These results demonstrate that an intrinsic departure from ideal (stable moisture) is to be expected even when care is taken to preserve and produce certain sample characteristics.

Compared with other reported litter characteristics, the commercial broiler litter used for this study appeared relatively uniform. Table 2 indicates a range of initial litter moisture from 11.5 to 31.2%, pH from 7.7 to 8.7, and total N content of 2.00 to 4.18%. The 11.5% moisture, a likely exception, resulted after a new management practice (rowing the litter between flocks) had been employed. Chamblee and Todd (2002) reported that litter moistures in Mississippi ranged from 19 to

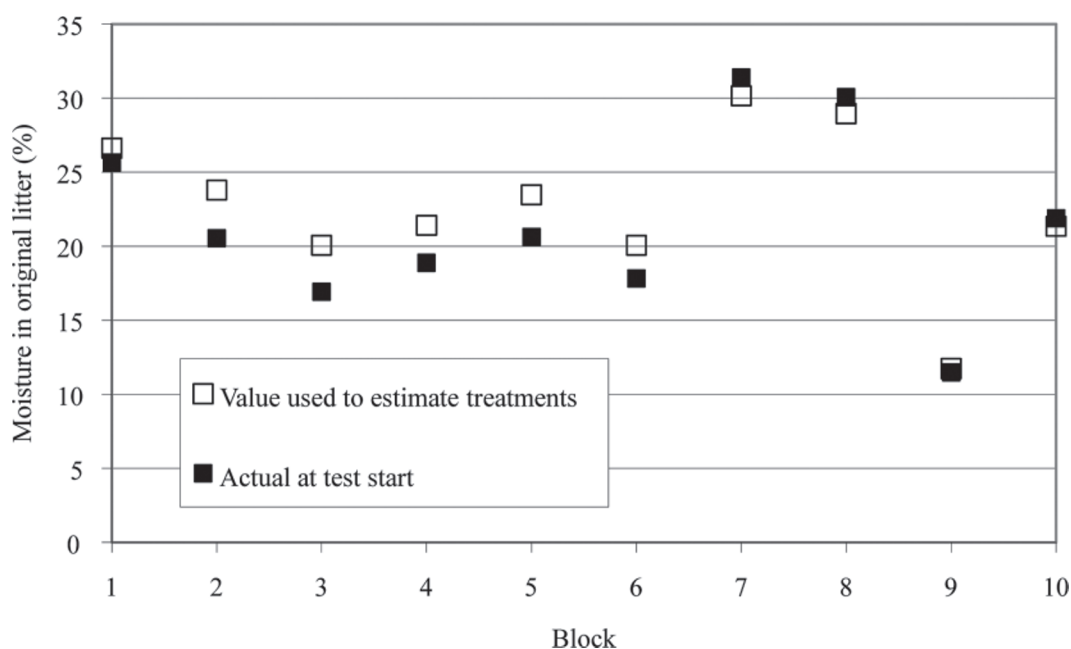


Figure 2. Storage effect on original litter moisture before the start of the test.

31% and the corresponding average N contents from 4 states were 2.9% (Mississippi), 3.1% (Alabama), 3.3% (Georgia), and 3.8% (Pennsylvania).

Another fundamental assessment of the system included changes in litter moisture content and pH over the 4-d experiment. As expected, greater temperatures led to increased moisture loss. Generally, there was an increase in pH over the course of the study, and the increase was associated with the higher moisture treatments, 35, 45, and 55% (data not shown). These changes are characteristic of the induced gradient moisture and temperature treatments in the CAT system. Other researchers have found similar results. An inability to control litter pH led Carr et al. (1990) to exclude it from an NH_3 volatilization model.

The overall agreement of the N mass balance for the study yielded 99.7%, with an SD of 8.8% between the initial and final N contents. Although the average agreement (99.7%) is acceptable, the SD (8.8%) is less than desirable. Plausible explanations for the deviation are given below. Table 3 shows that, in many instances, the N mass balance agreement for the individual incubators was greater than 100%. More N in the final litter sample is counterintuitive because the system measured N gaseous losses. The NH_3 loss was only about 2% of the total N, making the measurement of final N content in the litter samples responsible for the variability in agreement or percentage accounted for when final N > initial N. Possible reasons for this discrepancy include 1) heterogeneity of the litter, naturally varying up to 0.4% N between the initial and final samples; 2) inability to subsample effectively (e.g., bias based on particle size); and 3) analysis of very small quantities (40 mg).

NH_3 Volatilization

Ammonia volatilization (mg of N) from commercial broiler litter for temperatures 18.3 to 40.6°C, for initial moisture conditions (20.5 to 30.1%), and gradient moisture additions (25 to 55%) can be predicted using the following equation:

$$\text{Log predicted NH}_3 = b + (\beta_{\text{TL}} \times T) + (\beta_{\text{ML}} \times M) + (\beta_{\text{MTI}} \times T \times M) + (\beta_{\text{MQ}} \times M^2),$$

where b is the estimated intercept, T is the temperature (°C), M is the percentage of litter moisture, β_{TL} is the estimated linear temperature coefficient, β_{ML} is the estimated linear moisture coefficient, β_{MTI} is the estimated coefficient for the moisture $\times$ temperature interaction, and β_{MQ} is the estimated coefficient for the quadratic moisture response.

In the first ANOVA for the repeated measures, the 3-way interaction of temperature $\times$ moisture $\times$ day was not significant at the $\alpha = 0.05$ level ($P = 0.08$). All other 2-way interactions and main treatment effects were highly significant ($P < 0.0001$). A decision was made to assess the effects of temperature and moisture on each day as the most interesting and logical presentation of the results. The estimated coefficients for each day are given in Table 4. Practical use of the equations could include using house temperature and litter moisture values to predict litter NH_3 volatilization and would be the most useful to compare 2 temperatures at a single moisture content or to compare 2 moisture contents at a given temperature. Extrapolation to total flock NH_3 production is not recommended because of

Table 4. Probability of significance and estimated coefficients for predicting NH_3 volatilization from commercial broiler litter in the chamber acid trap system

Item ¹	Day of experiment			
	1	2	3	4
Probability of significance				
T	<0.0001	<0.0001	<0.0001	<0.0001
M	<0.0001	<0.0001	<0.0001	<0.0001
T $\times$ M	0.0231	0.0133	0.0004	0.0024
T $\times$ T	0.3906	0.9288	0.8962	0.6218
T $\times$ T $\times$ M	0.6166	0.5529	0.5028	0.8269
M $\times$ M	<0.0001	<0.0001	<0.0001	<0.0001
T $\times$ M $\times$ M	0.5200	0.9188	0.7635	0.6460
Model coefficient				
b	-0.3716	-0.5495	-0.04700	-0.01410
β_{TL}	0.01877	0.01583	-0.00026	-0.00399
β_{ML}	0.04019	0.05343	0.03728	0.03904
β_{MTI}	0.000485	0.000520	0.000843	0.000771
β_{MQ}	-0.00064	0.00078	-0.00070	-0.00071

¹T = temperature (°C); M = litter moisture content (%); b = estimated intercept; β_{TL} = estimated linear temperature coefficient; β_{ML} = estimated linear moisture coefficient; β_{MTI} = estimated coefficient for the moisture-temperature interaction; β_{MQ} = estimated coefficient for the quadratic moisture response.

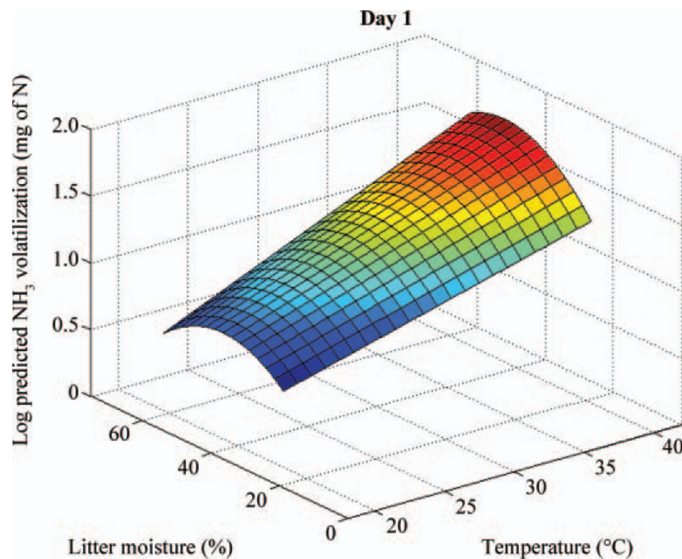


Figure 3. Day 1 predicted response of NH_3 volatilization from commercial broiler litter for moisture content ranging from 20 to 55% and temperatures ranging from 18 to 40°C. Color version available in the online PDF.

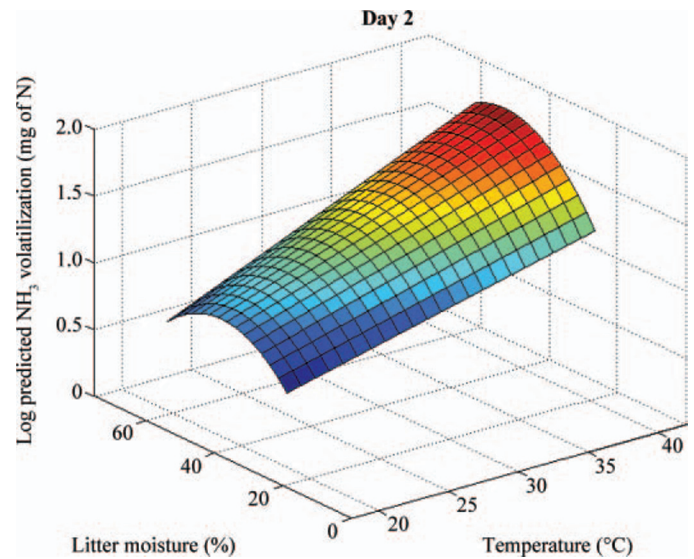


Figure 4. Day 2 predicted response of NH_3 volatilization from commercial broiler litter for moisture content ranging from 20 to 55% and temperatures ranging from 18 to 40°C. Color version available in the online PDF.

surface inconsistencies (e.g., presence of caked litter) within commercial houses. Validation of the equations at much larger surface areas would also be warranted.

The most notable results of the investigation can be seen in the surface response plots (Figures 3, 4, 5, and 6), developed from the prediction equations using coefficients in Table 4. Each successive figure was developed from an additional day of experimentation in which commercial broiler litter at gradient moisture content was enclosed in the CAT system. In Figures 3 to 6, the predicted response of the $\log_{10}$ NH_3 volatilization appeared as a parabolic cylinder (Weisstein, 2010). Ammonia evolution was consistently greater with increasing temperatures. The time response indicated that NH_3 generation increased from d 1 to 2, then decreased on d 3, and was further reduced on d 4, which was an expected phenomenon in the absence of additional fecal inputs and with decreasing moisture level of litter while enclosed in the CAT system. The effect of these factors places greater importance on earlier litter NH_3 generation results (d 1 and 2), where litter remains analogous to the initial treatment condition.

The consistent result from the NH_3 model was that maximum NH_3 volatilization occurs at an intermediate moisture level. For increasing moisture, NH_3 generation is maximized and then begins to decrease at a critical moisture point. The critical moisture content is temperature dependent. Table 5 provides the maximum NH_3 volatilization (values transformed from $\log_{10}$ notation) and the litter moisture at which NH_3 volatilization was maximized for d 1 and 2 of experimentation. Producers could use Table 5 as a guide to limit NH_3 generation by assessing litter temperature. For example, knowing that the litter temperature was 35°C, the producer would expect maximum NH_3 generation at litter moisture contents between 44.7 and 45.9%.

A depiction of the relative NH_3 emission rate from layer hen manure (Groot Koerkamp, 1998) divided the moisture content effects into 4 general levels relative to NH_3 generation and microbial activity: 0 to 5% moisture (a gradual, then sharp increase in NH_3 emission to approximately one-half of the maximum; moisture range associated with the destruction of microbes); 5 to 37% moisture (gradually increasing NH_3 release accompanied by moisture limited growth of microbes); 37 to 56% moisture (maximum NH_3 emission is apparent, but begins to decrease at approximately 50% moisture; optimal moisture range for the growth of microbes);

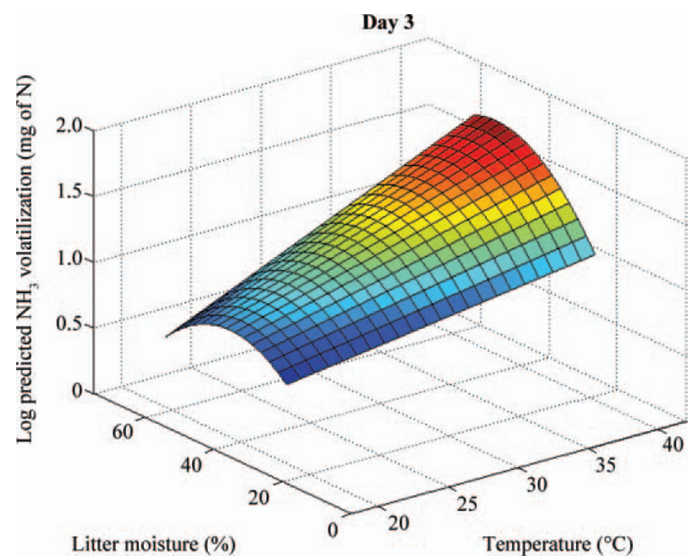


Figure 5. Day 3 predicted response of NH_3 volatilization from commercial broiler litter for moisture content ranging from 20 to 55% and temperatures ranging from 18 to 40°C. Color version available in the online PDF.

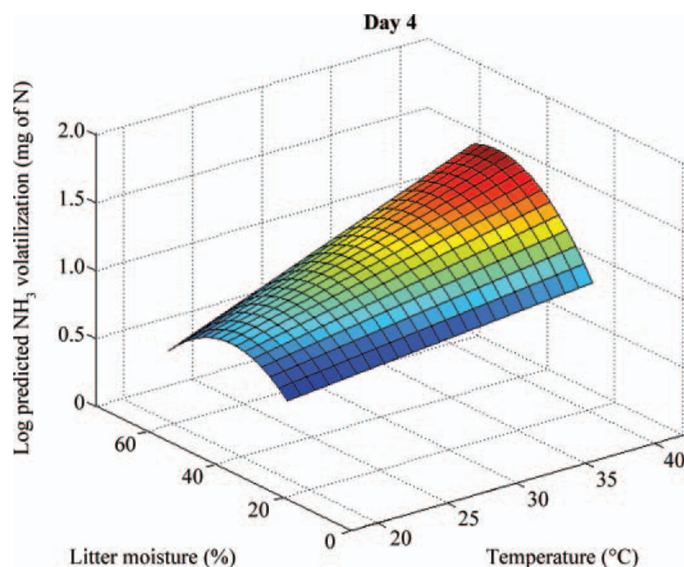


Figure 6. Day 4 predicted response of NH_3 volatilization from commercial broiler litter for moisture content ranging from 20 to 55% and temperatures ranging from 18 to 40°C. Color version available in the online PDF.

and 56 to 100% (declining NH_3 release under anaerobic conditions). The range for maximum NH_3 layer manure emission rate agrees quite well with the current study on commercial broiler litter (Table 5). For temperatures from 18.3 to 40.6°C, maximum NH_3 volatilization occurred between 38.3 and 46.8% moisture on d 1, between 40.4 and 47.8% moisture on d 2, between 37.6 and 51.1% moisture on d 3, and between 37.4 and 49.5% moisture on d 4.

In addition to moisture-related effects, microbial activity and specific populations would change with temperature. The temperatures investigated lie within the mesophilic temperature range (10 to 50°C; Madigan and Martinko, 2006). Most of the organisms in the litter are of fecal origin (34 to 37°C), so temperatures proximal to this range encourage optimal activity. The lower temperatures (near 18°C) cause an overall decline in activity (e.g., growth rate, replication, and enzyme activity slows) where urease activity and NH_3 production decrease (Muck, 1982). Although microbial activity and specific bacterial genera were not investigated in the current study, the influence of moisture and temperature on their activities was assumed. Inclusion of microbial analysis in future studies should enhance solutions for NH_3 mitigation.

Carr et al. (1990) investigated aerial NH_3 concentration from broiler chicken litter for a temperature range of 23.9 to 29.4°C, RH from 65 to 75%, 25 to 44% litter moisture, and a ventilation range from 2.2 to 20.5 air changes/h. The authors developed a $\log_{10}$ NH_3 equation with 3 linear terms (temperature, moisture, and ventilation) and 1 interaction term (temperature $\times$ moisture). At a constant flow rate (for comparison with the current study), Carr et al. (1990) indicated increasing NH_3 concentrations for the range of temperatures

Table 5. Maximum NH_3 volatilization at each temperature and the critical litter moisture contents

Temperature (°C)	Maximum NH_3 volatilized ¹ (mg of N)	Critical litter moisture content (%)
18.3	8.2 to 10.2	38.3 to 40.4
23.9	13.3 to 16.6	40.5 to 42.2
29.4	21.8 to 26.9	42.5 to 44.0
35.0	36.5 to 44.6	44.7 to 45.9
40.6	61.8 to 74.8	46.8 to 47.8

¹Range of values derived from d 1 and 2 of experimentation.

and moistures studied. Therefore, the effect of temperature is comparable with the results of Figures 3 to 6, but the authors did not exhibit a decline in aerial NH_3 near 40% litter moisture. The difference could be attributed to methodology because they intermittently measured NH_3 aerial concentration and disturbed the litter repeatedly vs. capturing volatilized gases in the CAT system without disrupting the litter surface.

Conclusions

Ammonia volatilization from broiler litter is dependent on litter moisture content and temperature. Increasing the temperature causes greater NH_3 volatilization. However, NH_3 generation reaches a maximum at intermediate (critical) litter moisture. Volatilization of NH_3 from broiler litter over the range of temperatures (18.3 to 40.6°C) and litter moisture contents (20 to 55%) found in commercial houses was predicted using a log NH_3 equation. The log response surface was a parabolic cylinder, indicating that maximum NH_3 production was between 37.4 and 51.1% litter moisture. Maximum NH_3 volatilization at the low temperature (18.3°C) corresponded to 37.4 to 40.4% litter moisture, whereas the maximum at the upper limit temperature (40.6°C) occurred between 46.8 and 51.1% litter moisture, both depending on the elapsed time during the experiment. Above these critical moisture levels for a given temperature, NH_3 generation decreased. It is not known whether the reduction was due to anaerobic conditions within the litter or to partitioning of NH_3 into water-filled pore spaces, a combination of the 2, or other additional dynamics within the litter. This research defines intermediate critical moisture levels in broiler litter where NH_3 is maximized, providing target areas for researchers and the poultry industry to develop management scenarios to reduce NH_3 production from litter.

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